Lab 11
Display Student ID & Birthday

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Contents of this slide are provided by NTOU VLSI Lab601

Objectives

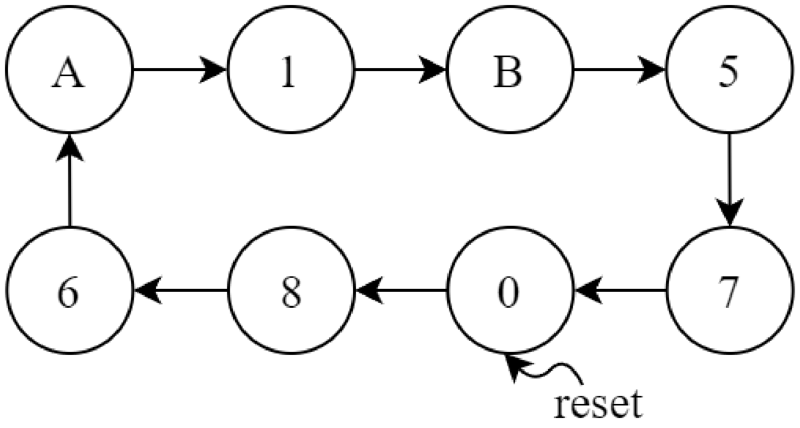
- Display Student ID
- Display Birthday

LAB 1 – Display Student ID (1/5)

- Design a Finite State Machine (FSM) that outputs digits of an encoded student ID one by one.
- Rules to encode the student ID
  - When a repeated digit is encountered, replace the first occurrence with ‘A’, the next with ‘B’, and so on.
  - The last occurrence of each repeated digit remains unchanged.
    - For example, 00957001 becomes **AB**957**C**01.
  - When there are multiple sets of repeated digits in the student ID, replace the repeated digits with letters A ~ F in the order they appear.
    - For example, 00957007 becomes **AB**95**CD**07.
  - When there are letters already present in the student ID, skip the original letters.
    - For example, 0086B002 becomes **AC**86**BD**02.

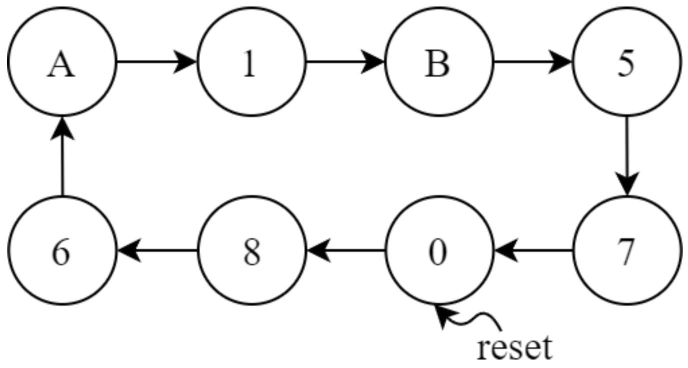
LAB 1 – Display Student ID (2/5)

- Signal settings
  - Input: clk and reset
  - Output: student_id[3:0]
- State diagram: Assume 01057086 → A1B57086



LAB 1 – Display Student ID (3/5)

- State table: Assume 01057086 → A1B57086



Current status (t)					Next status ($t+1$)				
ID	$W(t)$	$X(t)$	$Y(t)$	$Z(t)$	$W(t+1)$	$X(t+1)$	$Y(t+1)$	$Z(t+1)$	ID
0	0	0	0	0	1	0	0	0	8
1	0	0	0	1	1	0	1	1	B
	0	0	1	0	X	X	X	X	
	0	0	1	1	X	X	X	X	
	0	1	0	0	X	X	X	X	
5	0	1	0	1	0	1	1	1	7
6	0	1	1	0	1	0	1	0	A
7	0	1	1	1	0	0	0	0	0
8	1	0	0	0	0	1	1	0	6
	1	0	0	1	X	X	X	X	
A	1	0	1	0	0	0	0	1	1
B	1	0	1	1	0	1	0	1	5
	1	1	0	0	X	X	X	X	
	1	1	0	1	X	X	X	X	
	1	1	1	0	X	X	X	X	
	1	1	1	1	X	X	X	X	

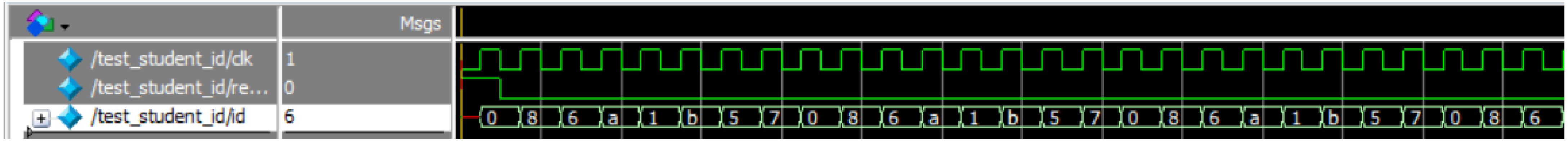
LAB 1 – Display Student ID (4/5)

- Testbench

```
module test_student_id;

    logic clk, reset;
    logic [3:0] id;

    student_id id1(
        .clk(clk),
        .reset(reset),
        .id(id)
    );

    always #5 clk = ~clk;

    initial begin
        clk = 0; reset = 1;
    #10 reset = 0;
    #500 $stop;
    end

endmodule
```

LAB 1 – Display Student ID (5/5)

- Waveform: Assume 01057086 → A1B57086



- When downloading to the DE0 board, the clock must pass through a frequency divider before use.
  - Use freq_div_23 designed in Lab 9
- The output student_id should be connected to the seven-segment display.

```
freq_div_23 fre_23(
    .clk        (CLK),
    .reset      (~BTN[0]),
    .clk_out    (clk_out)
);

student_id id1(
    .clk(clk_out),
    .reset(~BTN[0]),
    .student_id(id)
);

seven_segment s0(
    .in     (id),
    .out    (HEX0)
);
```

LAB 2 – Display Birthday (1/3)

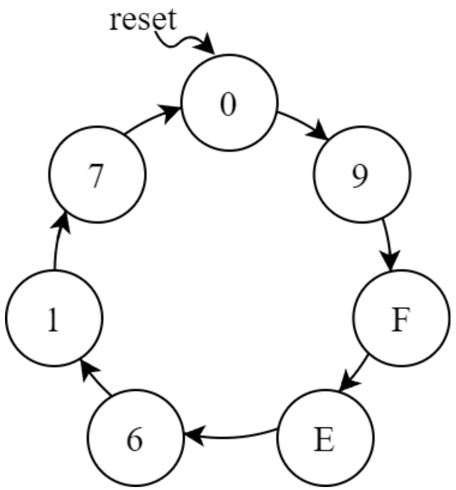
- Design a Finite State Machine (FSM) that outputs digits of a birthday in the format of YYYMMDD one by one.
  - For example, if the birthday is June 17, 2010, it will be formatted as 0100617.
- When a repeated digit is encountered, the second occurrence of the repeated digit is replaced with ‘F’, the third one is replaced with ‘E’, and so on.
  - For example, 0990617 becomes 09FE617. The first occurrence of a repeated digit remains unchanged.

LAB 2 – Display Birthday (2/3)

- Rules for modifying repeated digits in the birthday:
  - The first occurrence of a repeated digit remains unchanged, while all previous occurrences are replaced sequentially with the letters F, E, D, C, B, A.
    - For example, 0981217 becomes 09811F7.
  - When there are multiple sets of repeated digits in the birthdate, replace them in order of appearance using letters from F to A.
    - For example, 0990617 becomes 09FE617.
- If you are unsure how to convert your birthday, please ask the TA for help.

LAB 2 – Display Birthday (3/3)

- Signal settings
  - Input: clk and reset
  - Output: birthday [3:0]
- State diagram: Assume 0990617 → 09FE617



- When downloading to the DE0 board, the clock must pass through a frequency divider before use.
  - Use freq_div_23 designed in Lab 9
- The output should be connected to the seven-segment display.